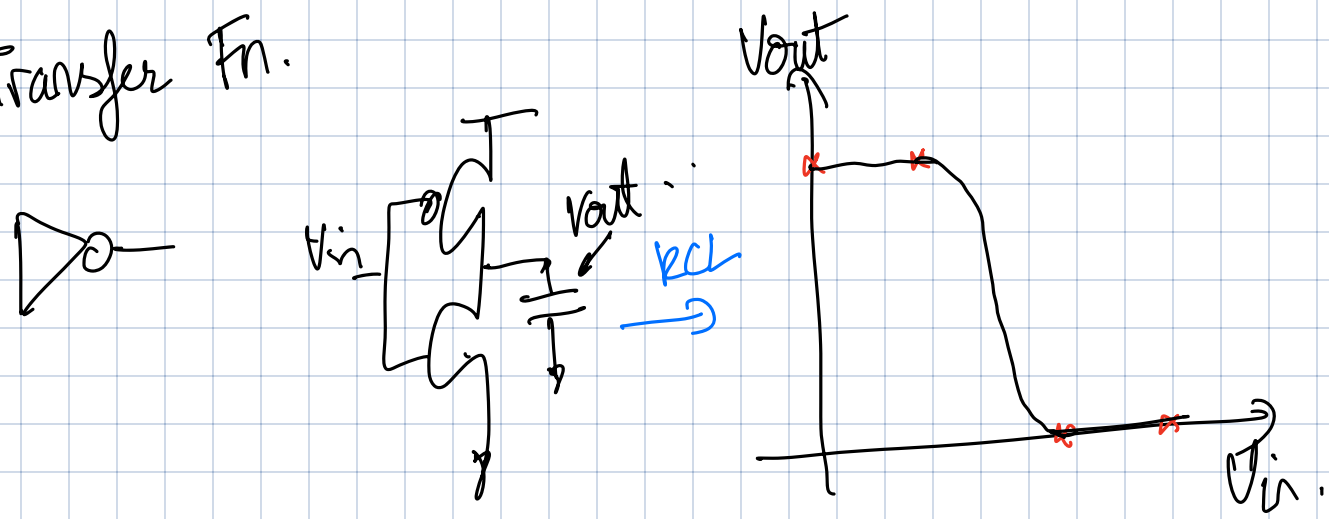
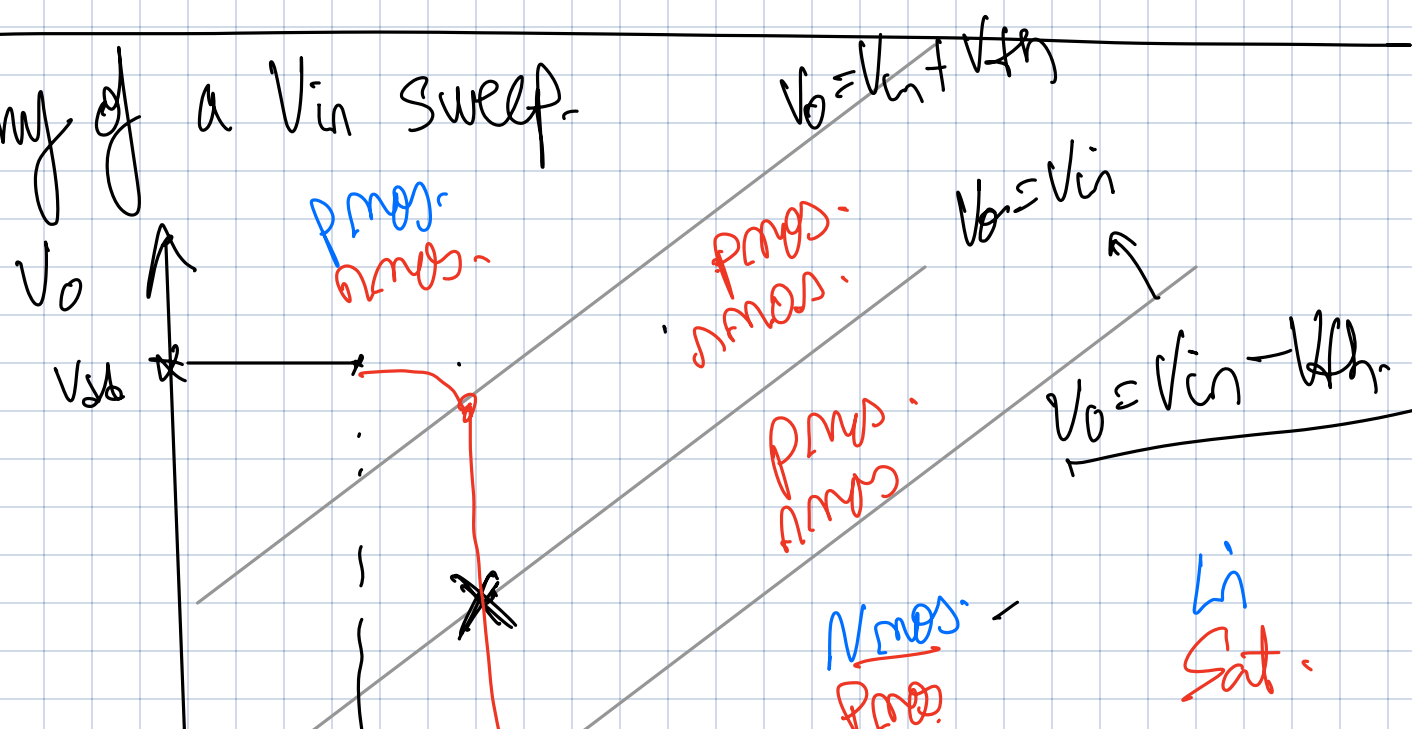
1. DC transfer function
2. Regions of operation (when am I a resistor and when am I a current source)
3. Thought exercises (vth differences, vdd<2vth, flipped inverter, negative vth!)
4. Dc noise tx fn
5. Cascaded inverters….
6. Schmitt triggers
7. Gain, nonlinearity and logic

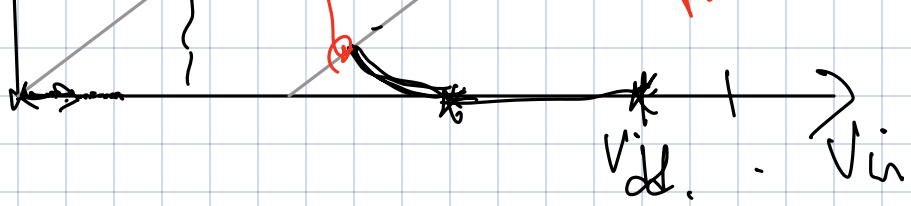
1. X-talk
2. Setup
3. Level1 glitch
4. level2 glitch
5. Miller coupling

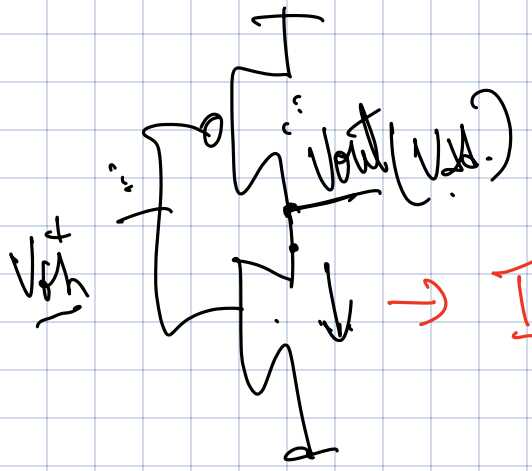
DC Transfer Fn.



Anatomy of a Vin sweep.

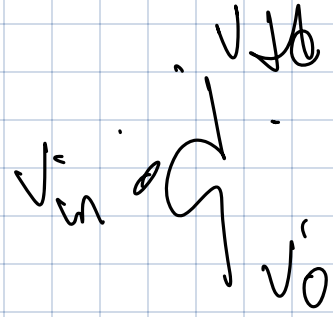






$\rightarrow I_n \approx \frac{1}{2}\beta V_{gs}'^2$

When is Pmos in sat?



All

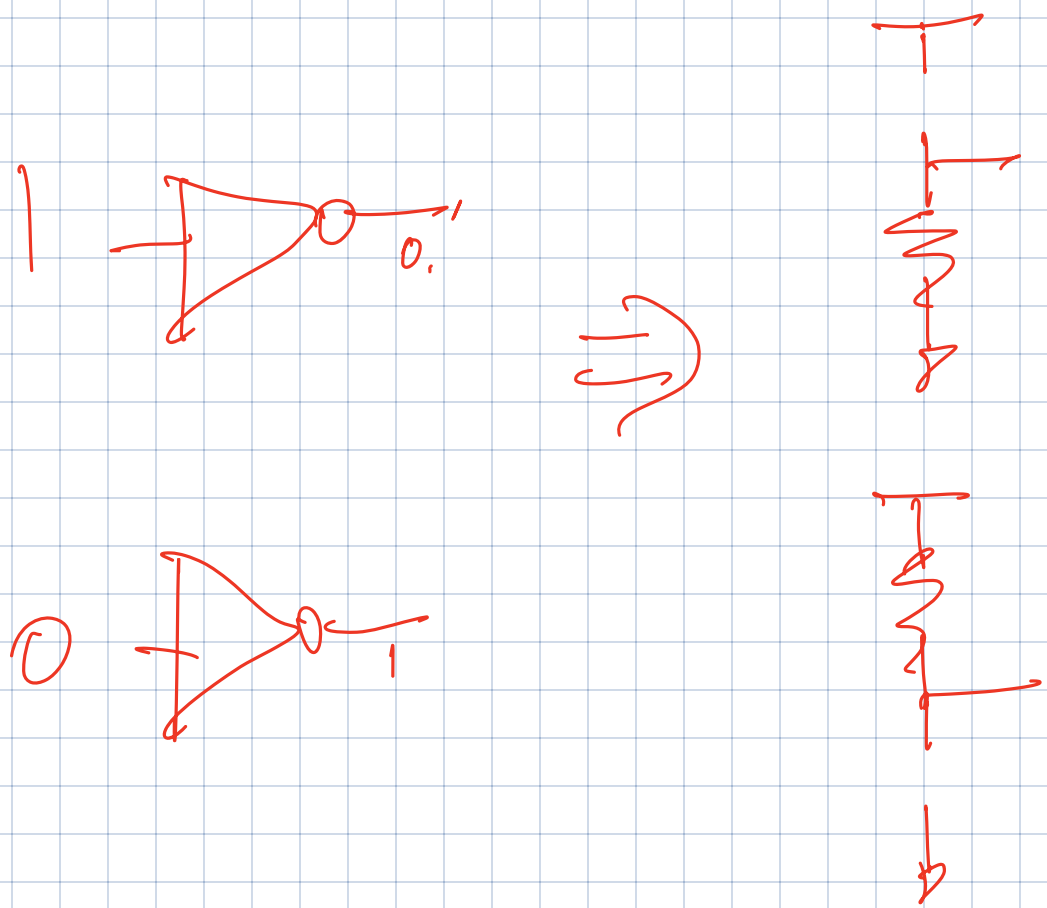
$$V_{dd} - V_{in} - V_t < V_{dd} - V_o$$

$$V_o < V_{in} + V_{th}$$

When is nmos in sat?

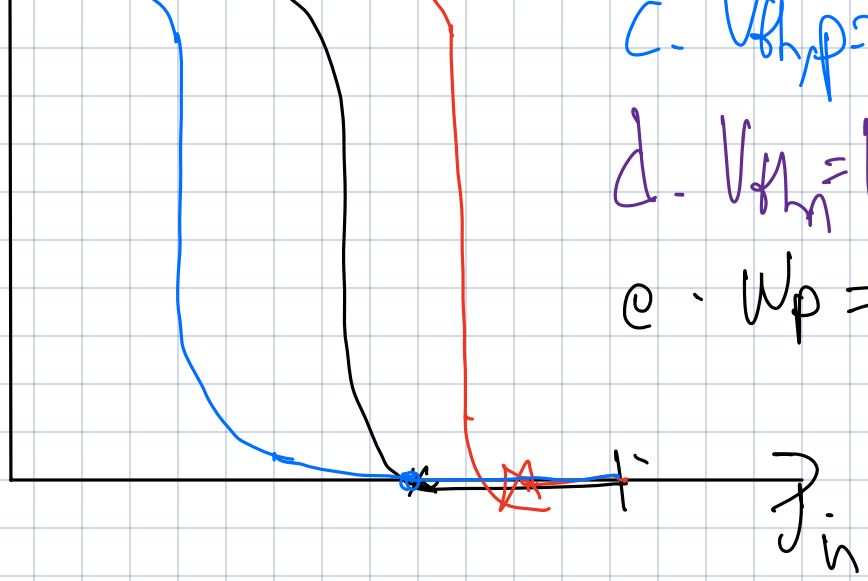
$$\left(V_o > V_{in} - V_{th}\right)$$

Takeaway for coupling:



a. $V_{thn} = 0.2 = V_{thp}$.

b. $V_{thn} = 0.4$, $V_{thp} = 0.2$

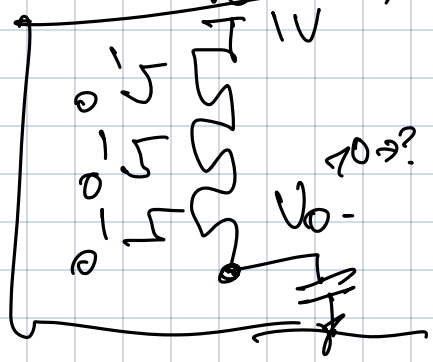


c. $V_{th,p} = 0.4$, $V_{th,n} = 0.2$

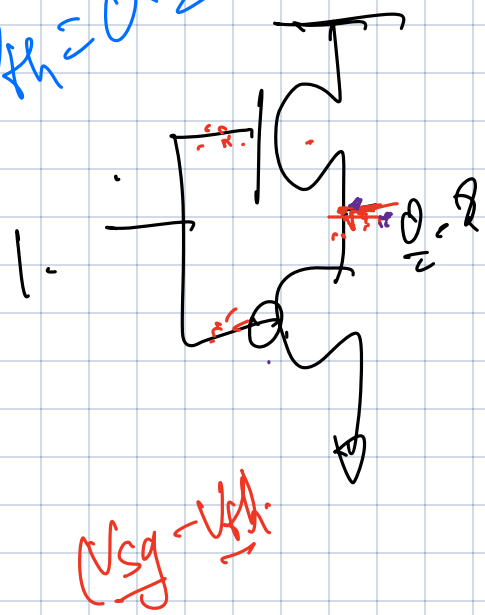
d. $V_{th,n} = V_{th,p} = 0.6$, $V_{dd} = 1V$

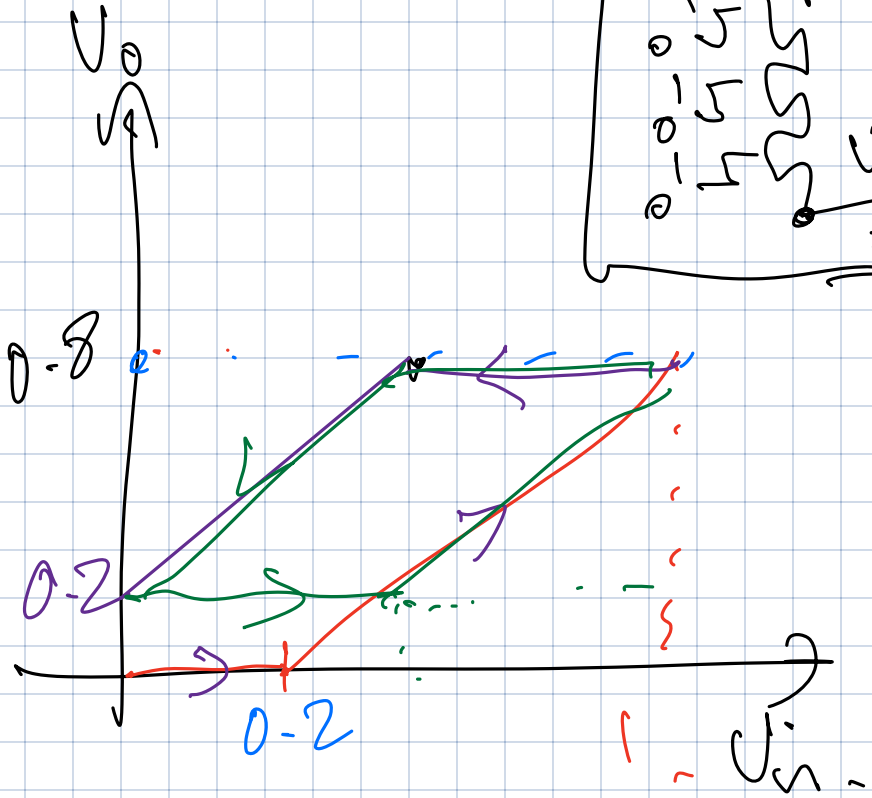
e. $W_p = 6W_n$?

Common Qn. Teaser:



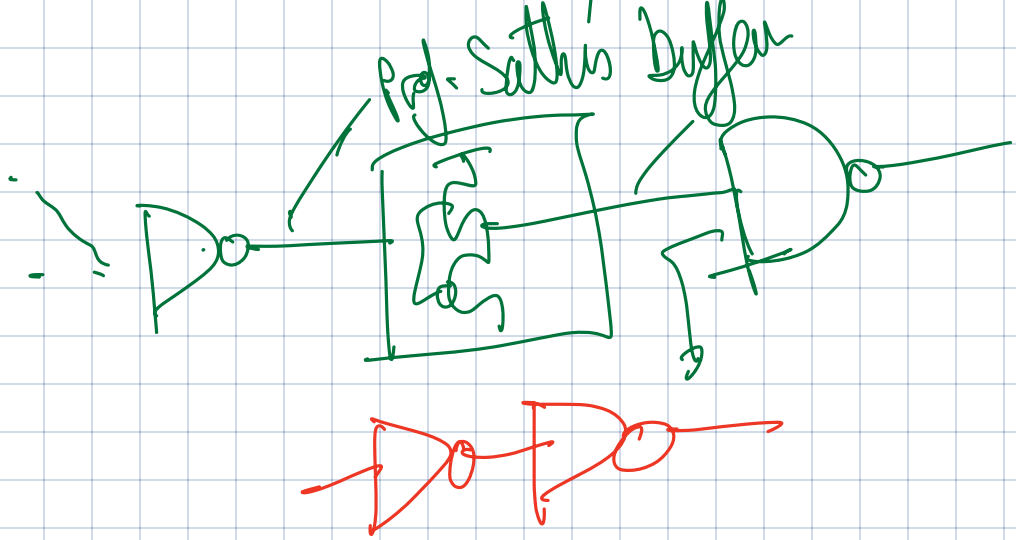
$V_{th} = 0.2$

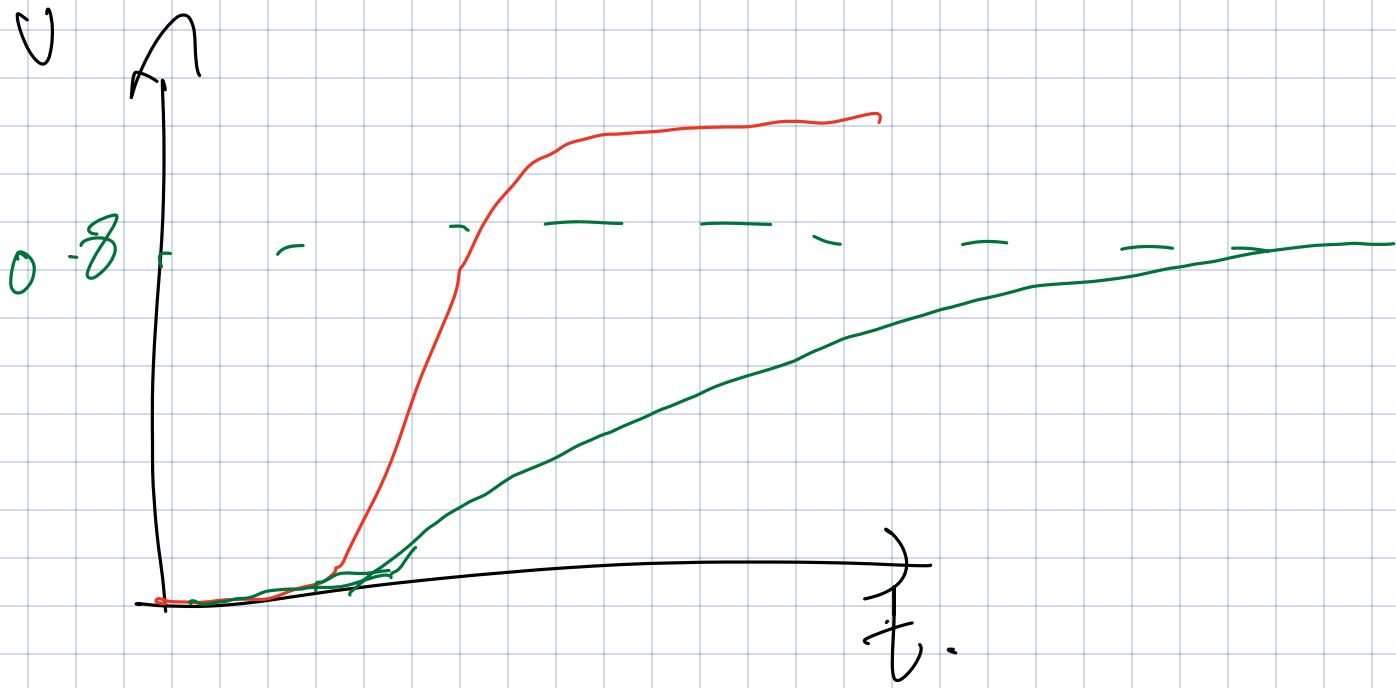


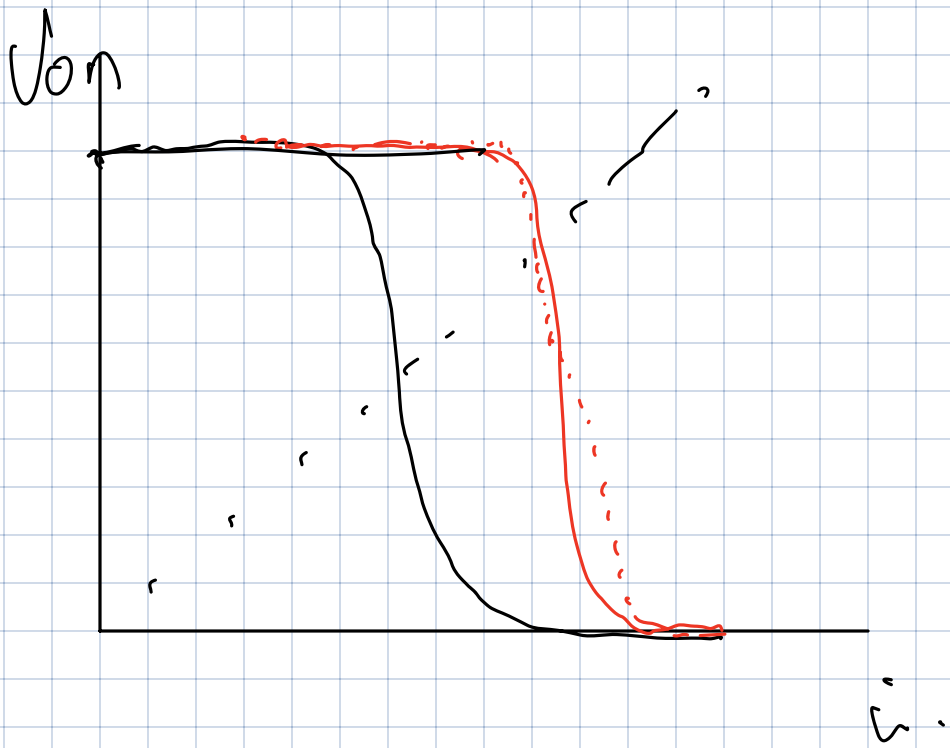


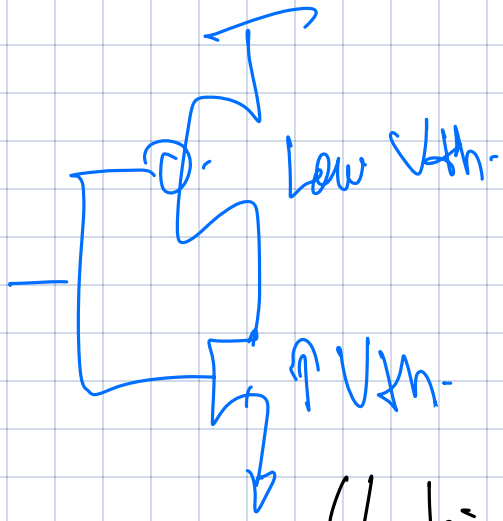
Why this buffer is lousy ?

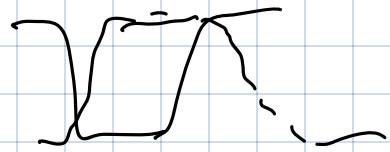
1. No 0V → V_{dd} swing.


Prof. Sathis Buffer

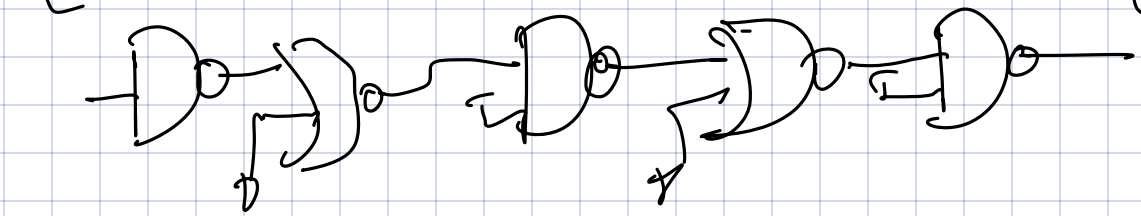

V
0.8
t


Von
i





Let "x" be your "robustness" to noise @ the input in V-units



Noise: Should I $\min[x_1 + x_0]$?

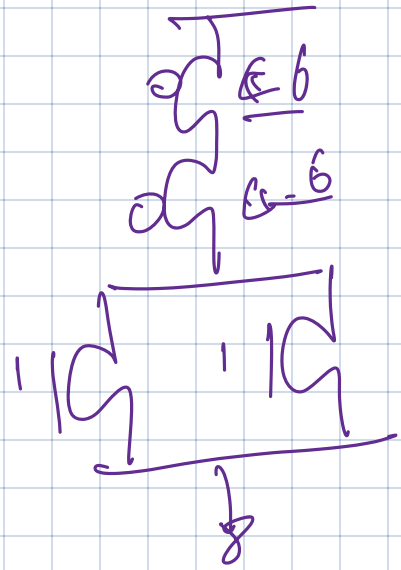
$x_1 = 0.2V$
$x_0 = 0.2V$.

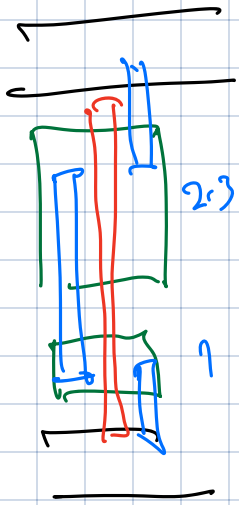
$x_1 = 0.3V$
$x_0 = 0.15V$.

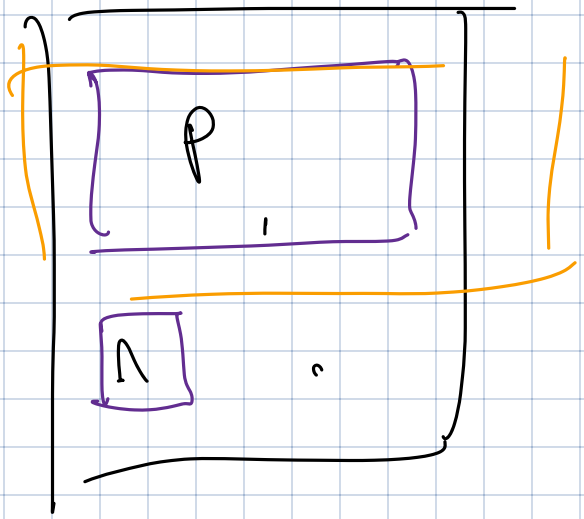
You should be focussed on: $\max(\min(x_0, x_1))$

In 65nm

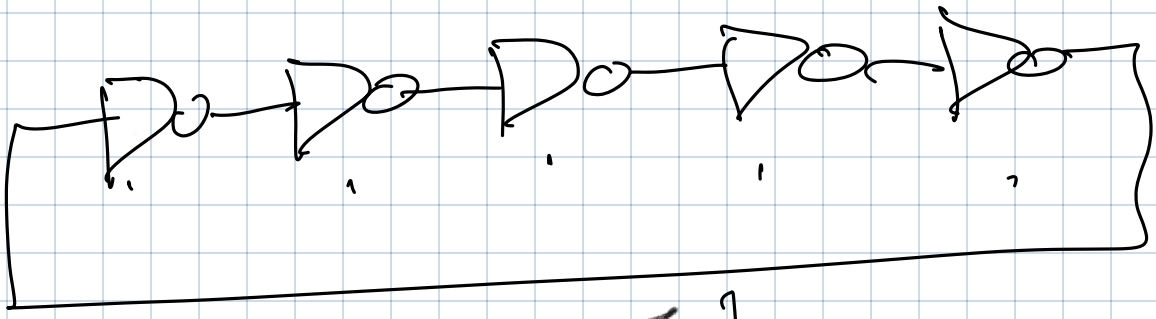
$\beta \approx 2.3$



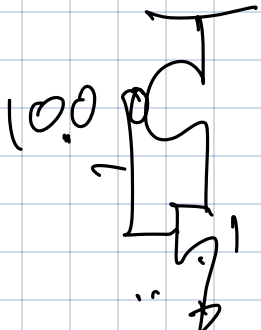




$\beta = \beta_0$

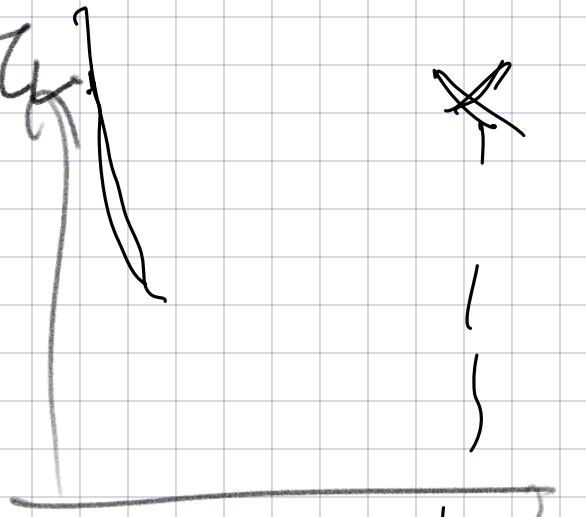


$\tau_1 + \tau_2$



τ

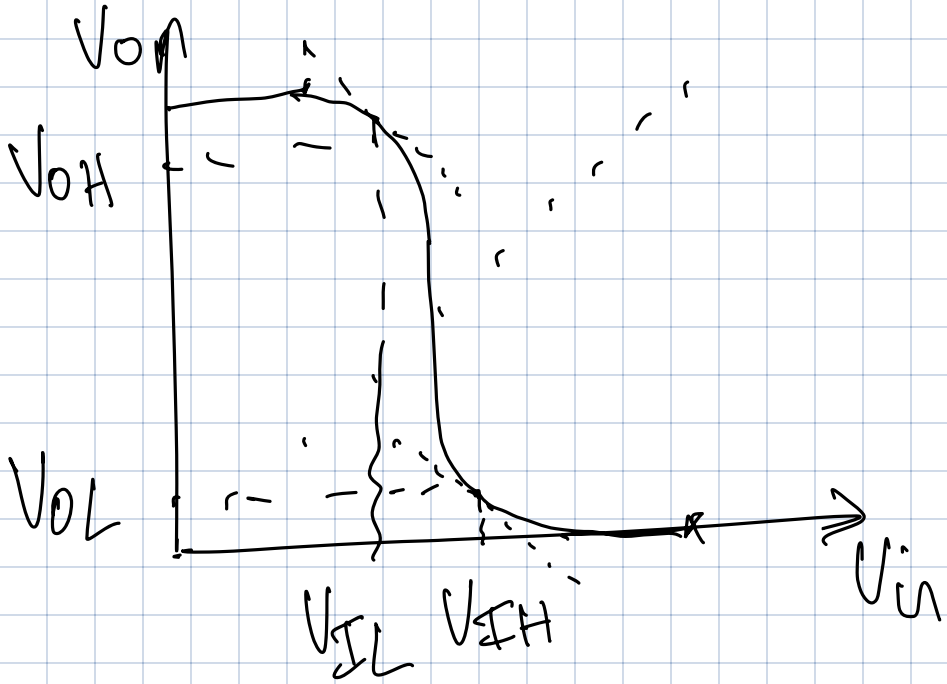
τ_{100}

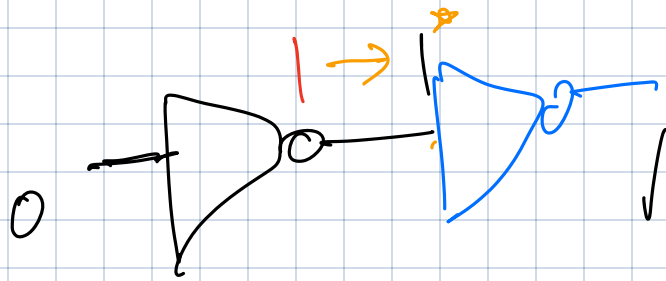


Noise Margins (Inverter)

Relies on:

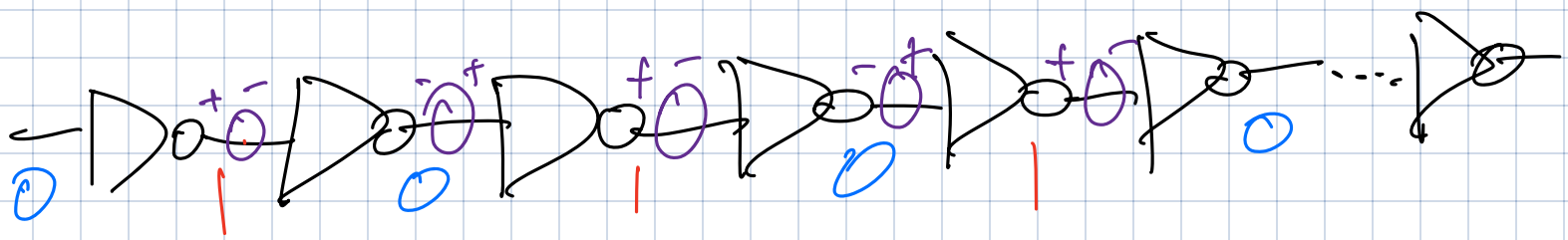
max(min(NM0, NM1))



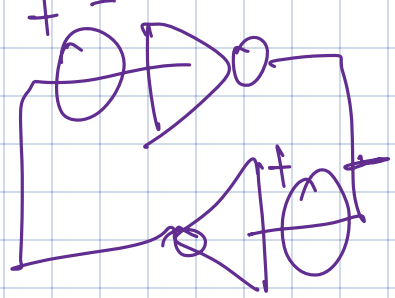


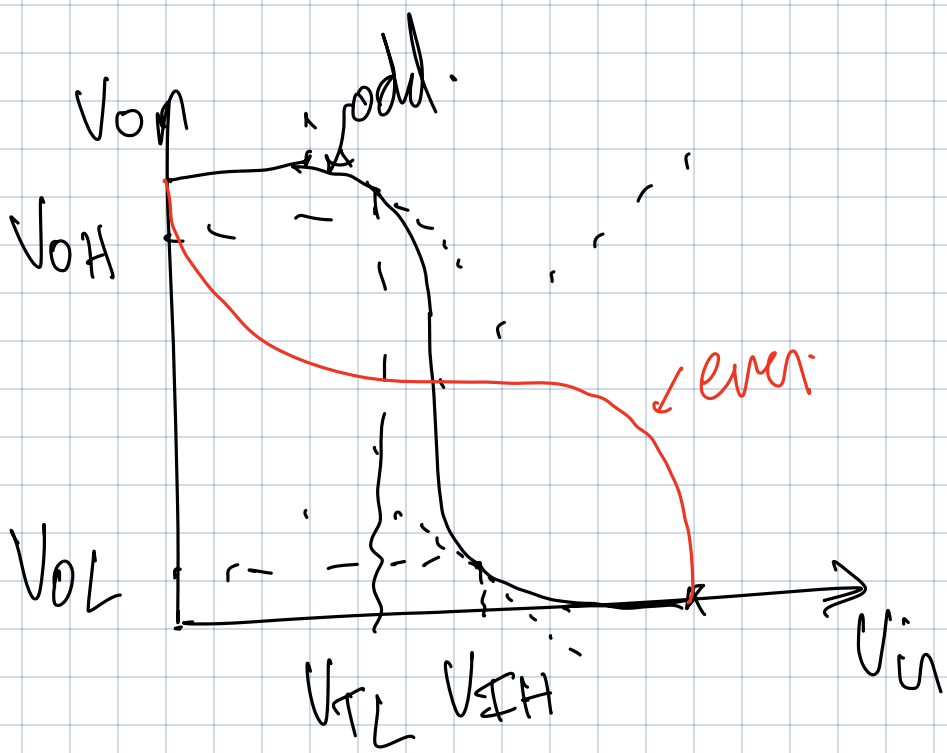
$$NM0 = V_{IL} - V_{OL}$$

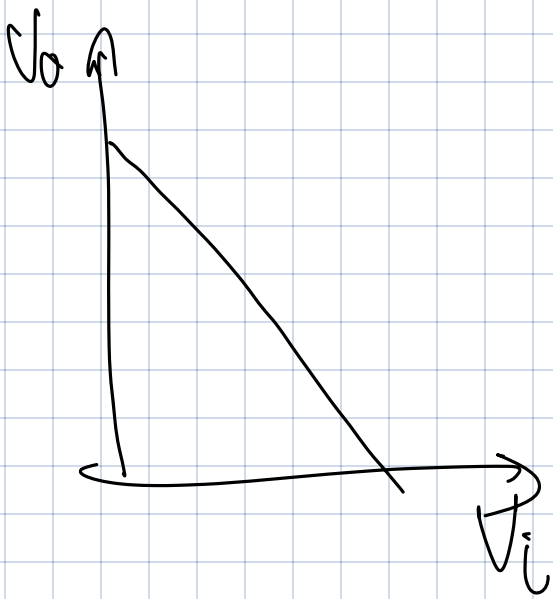
$$NM1 = V_{OH} - V_{IH}.$$



$\Downarrow$

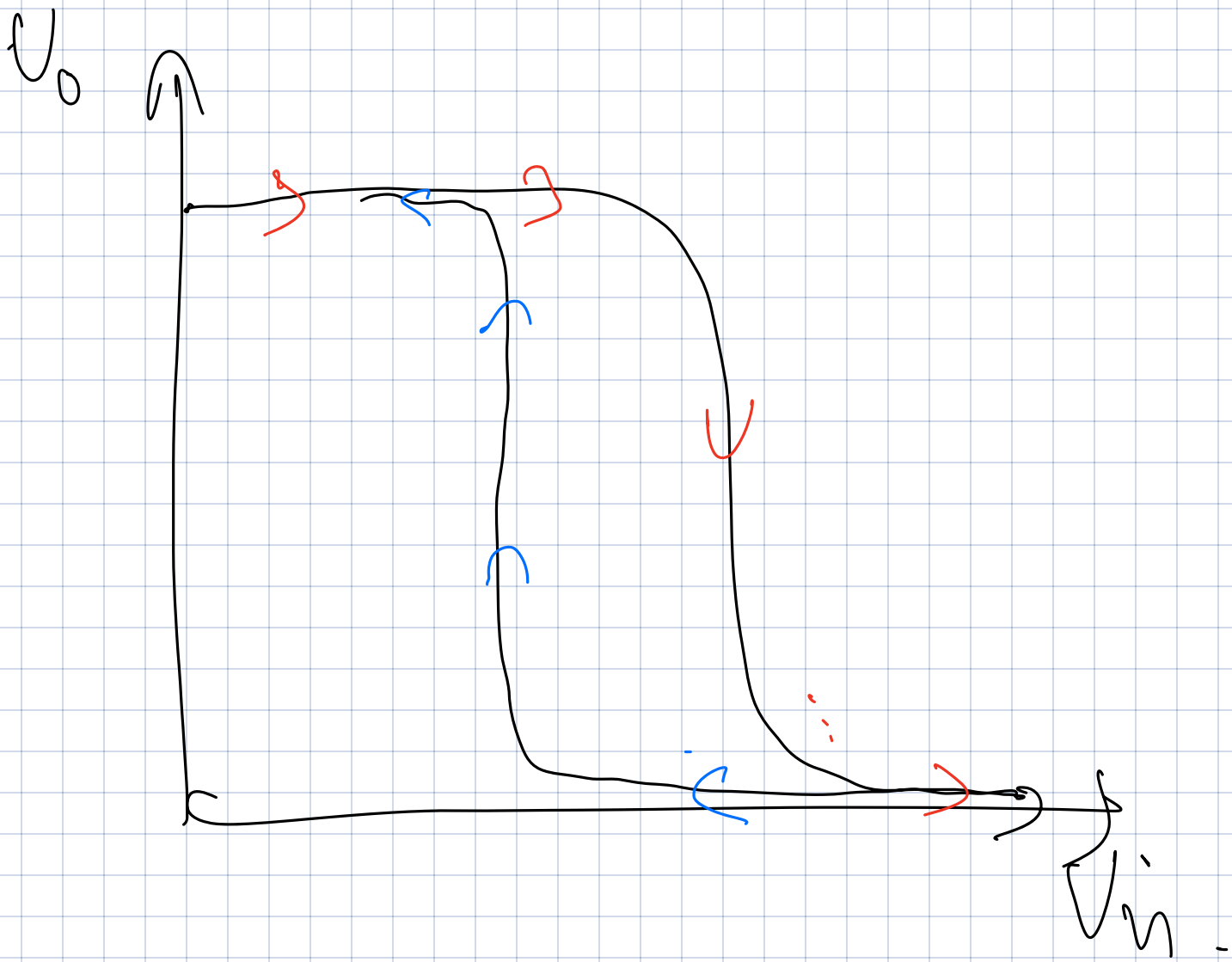


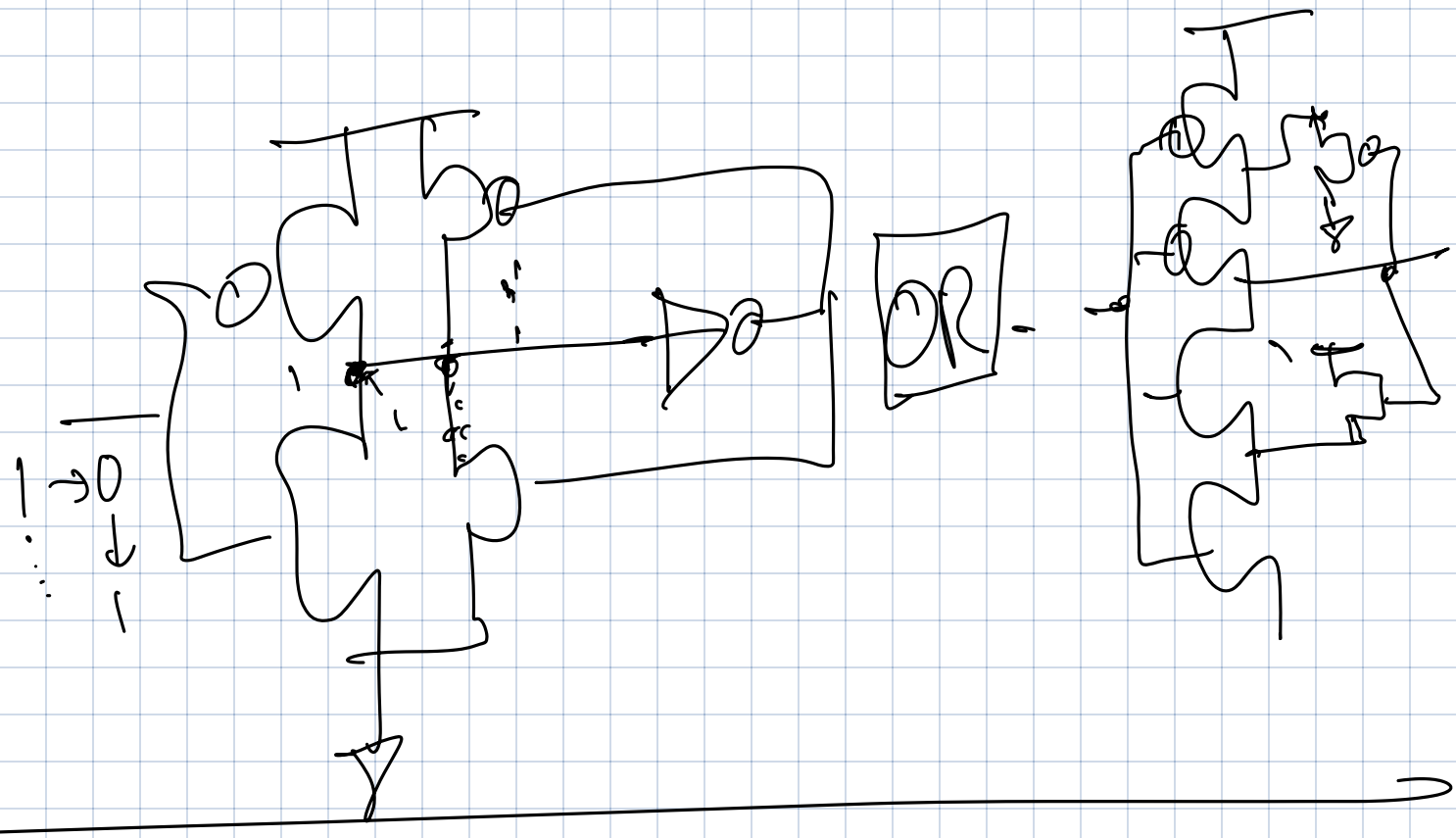




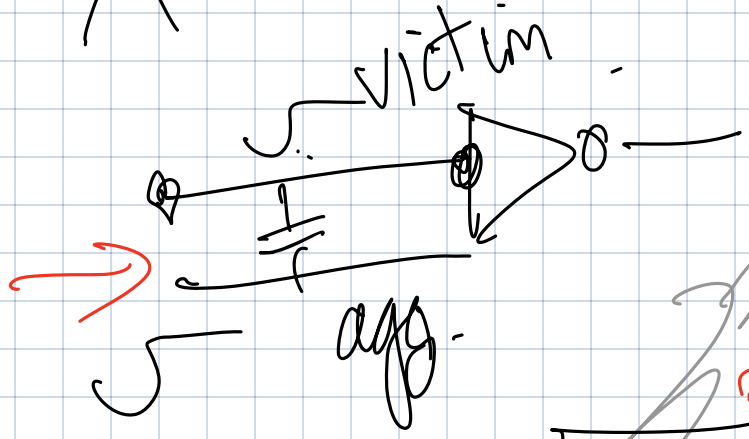
"Bad" logic

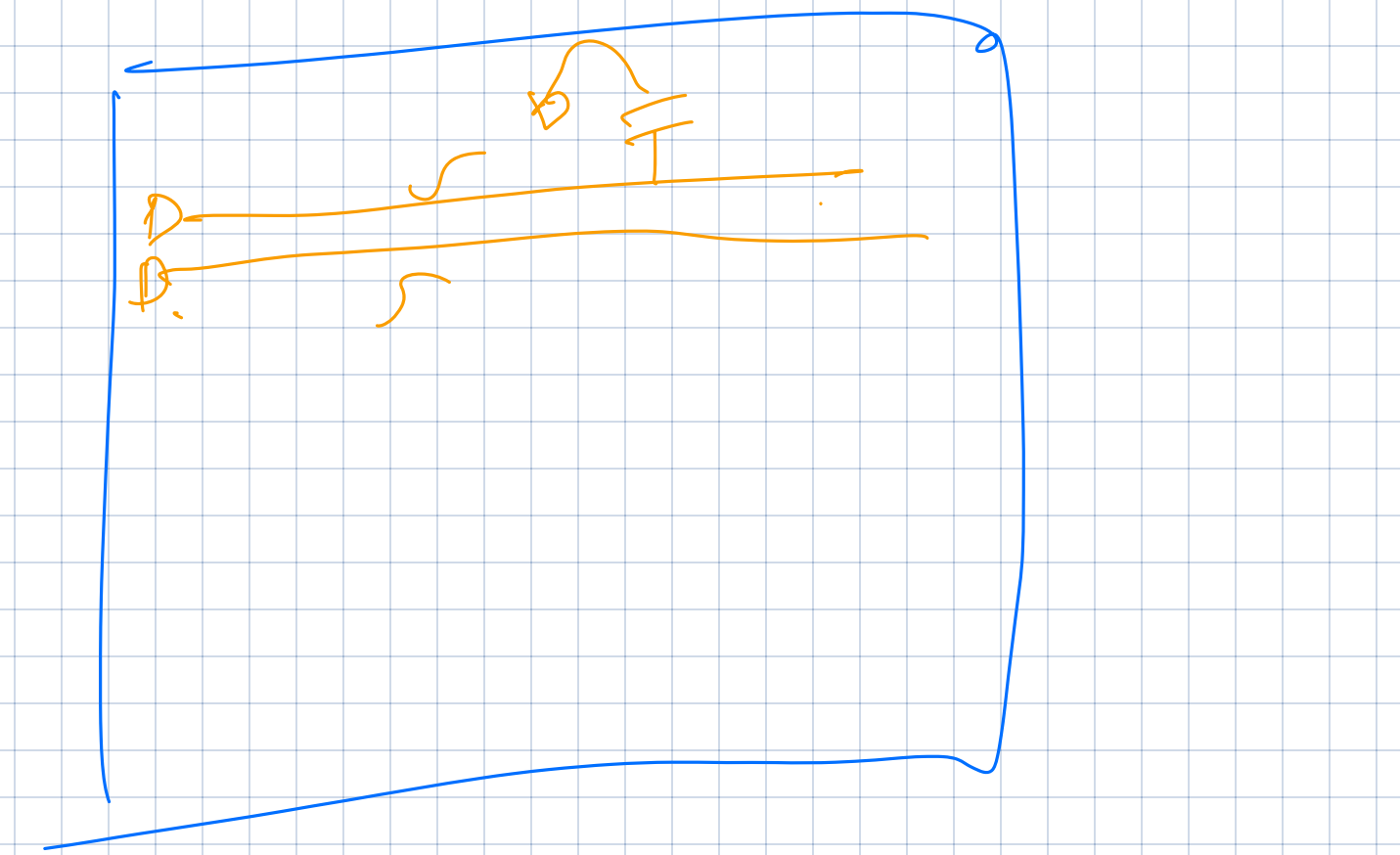
No gain/non-linearity

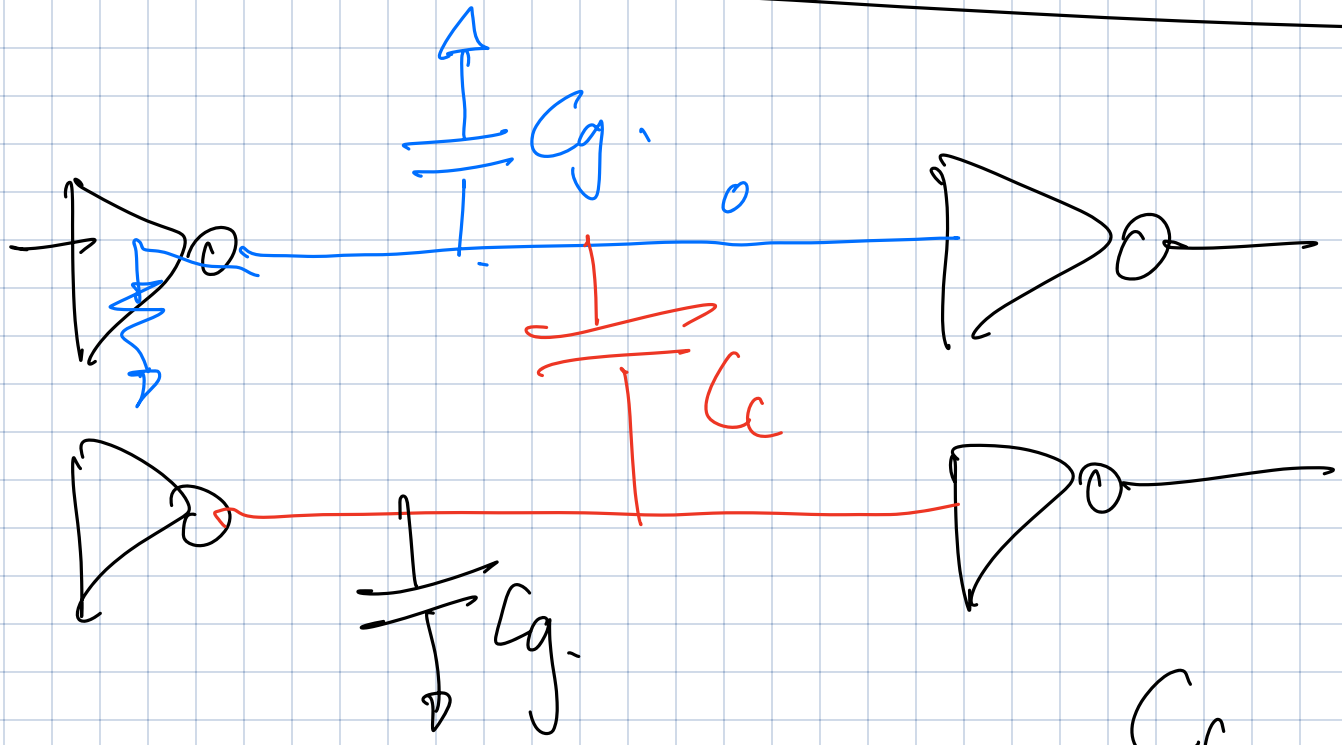

V_o
V_{in}

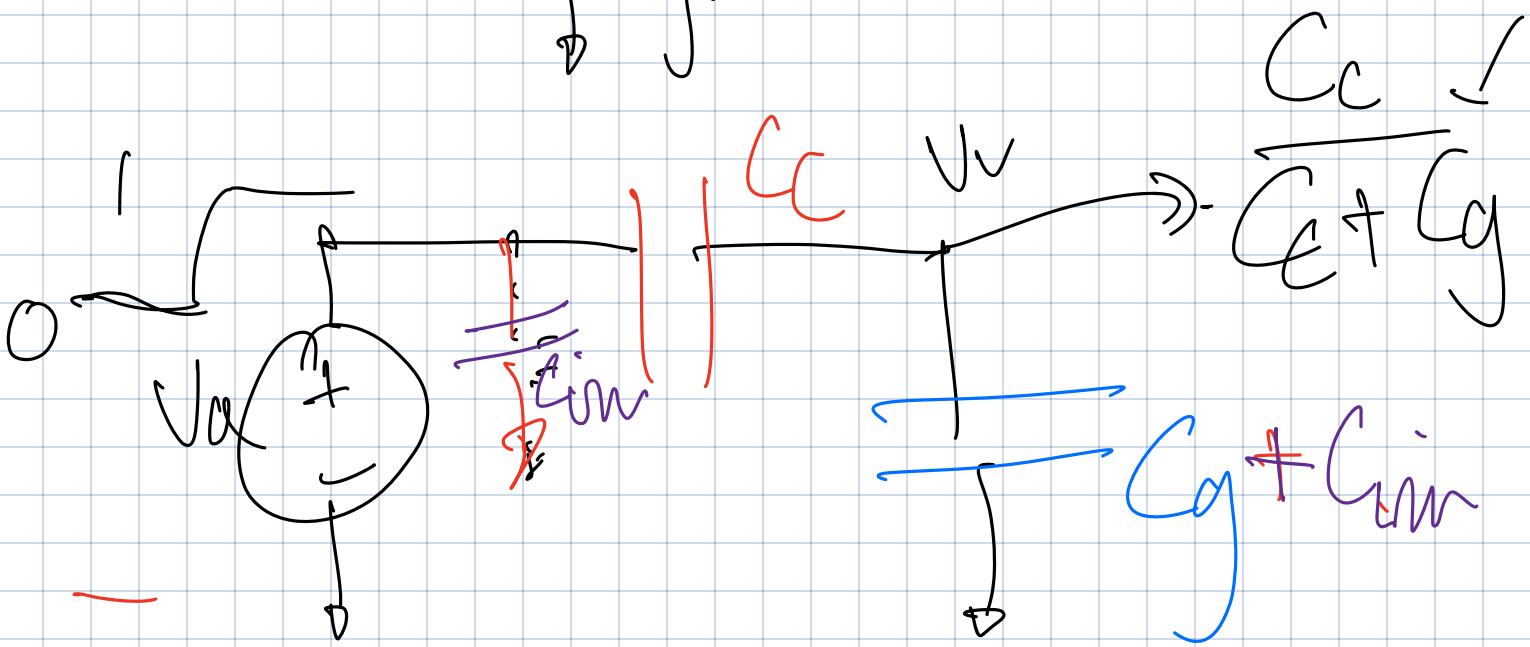

1 → 0
OR

X - Talk.

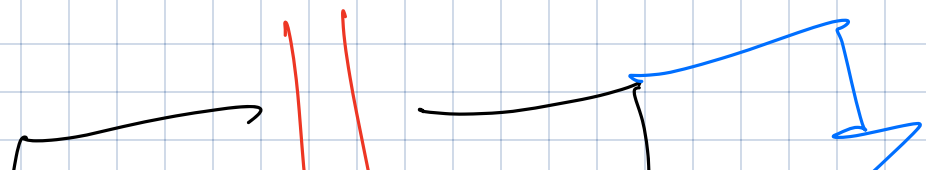


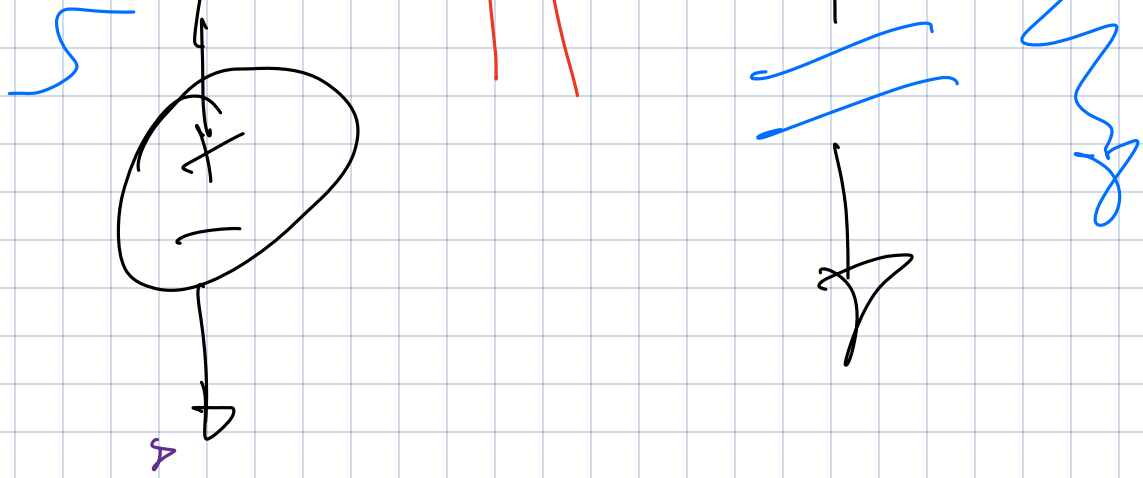




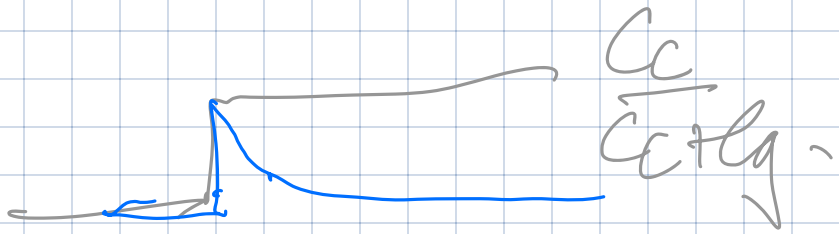


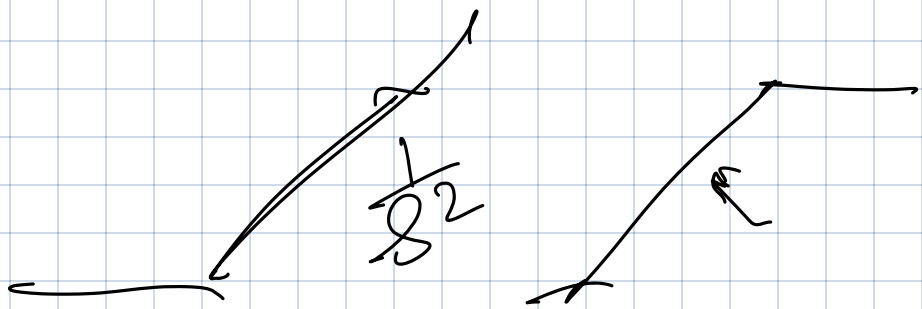
Above model has a glaring (& convenient) omission.





Va
1/s


Cc
Cc+Cg


1/s^2